<table>
<tr><td>1</td><td>You are given N bank transactions.
Each transaction contains:
Sender
Receiver
Amount
Timestamp
A transaction is considered fraudulent if:
The Sender, Receiver, and Amount are the same as a previous transaction.
The difference between their timestamps is ≤ 60 seconds.
Input:
5
ANU JON 200 1000
ANU JON 200 1050
RAM SAM 300 2000
ANU JON 200 2000
RAM SAM 300 2050
Output:
ANU JON 200 1050
RAM SAM 300 2050</td></tr>
<tr><td>2</td><td>A gym offers membership plans:
<table>
<tr><td>Months</td><td>Cost</td></tr>
<tr><td>12</td><td>₹15000</td></tr>
<tr><td>9</td><td>₹12000</td></tr>
<tr><td>6</td><td>₹9000</td></tr>
<tr><td>3</td><td>₹5000</td></tr>
<tr><td>1</td><td>₹2000</td></tr>
</table>
Given N months, find the minimum cost to purchase membership.
You can buy multiple plans.
Input:
15
Output:
20000</td></tr>
<tr><td>3</td><td>A hot air balloon has a maximum weight capacity X kg.
You are given the weights of N people. Find the maximum number of people that can travel in the balloon such that the total weight does not exceed X. To maximize the count, choose the lightest people first.

Input:
6 (People)
4 2 7 1 3 5
10 (Value of X)
Output: 4</td></tr>
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4	Given an array of integers (positive, negative, and zero), find the maximum product that can be obtained from any non-empty subset of the array. **Input** [-1, -1, -2, 4, 3] **Output** 24 Explanation: (-1) × (-1) × 4 × 3 = 12 (-1) × (-2) × 4 × 3 = 24 ← Maximum
5	A laptop needs a minimum charge X to work. Count how many laptops can work (i.e., charge ≥ X). X = minimum charge required for a laptop to work Array A [] containing charge percentages of laptops **Input**: 30 10 40 50 20 35 **Output**: 3
6	You are given an integer n representing a number of people initially in happy state In each iteration • 70% happy becomes Sad • 30% of happy remains happy • 50% sad remains Sad • 50% sad becomes Happy **Input**: 100 3 **Output**: Happy: 41.2, Sad: 58.8
7	Given an array of N integers (which may contain positive, negative, or zero values), find the maximum possible sum of any contiguous subarray. A subarray is a continuous part of the array. **Input**: 8 -2 -3 4 -1 -2 1 5 -3 **Output:** 7
8	Given an array of N integers, find: • The element with the minimum frequency. • The element with the maximum frequency. If multiple elements have the same frequency: • For maximum and minimum frequency, return the element that appears first in the array. **Input**: 8 1 2 2 3 3 3 4 4 **Output**: Minimum Frequency Element = 1 Maximum Frequency Element = 3
9	There are N people standing in a circle, numbered from 1 to N. Starting from person 1, every K-th person is eliminated from the circle. After a person is removed, counting continues from the next remaining person. This process continues until only one person remains.

	Your task is to fin1d the number of the last remaining person. **Input**: 8 4 **Output**: 6
10	You are given a **sorted array of integers** of size N and a target value X. (Use Binary Search only) Your task is to find: 1. The index of the **first occurrence** of X in the array. 2. The index of the **last occurrence** of X in the array. If X is not present in the array, return -1. **Input**: 8 1 2 2 2 3 4 4 5 2 **Output**: 1 and 3
11	You are given a string consisting of lowercase English letters. Encode the string as follows: • For characters at **even indices** (0-based), shift the character **forward by 2** positions in the alphabet (+2). • For characters at **odd indices**, shift the character **backward by 1** position in the alphabet (-1). • Alphabet wrapping is allowed: ○ 'y' + 2 = 'a' ○ 'z' + 2 = 'b' ○ 'a' - 1 = 'z' Return the encoded string. **Input**: abcdef **Output**: caecge
12	You are given two integers N and S. Starting from S, find the first prime number greater than or equal to S. Then repeatedly: • Take the next two prime numbers. • Find their sum. • Continue this process until the sum becomes greater than or equal to N. Print the final sum obtained. **Input**: N = 50 S = 10 **Output**: 52 **Explanation**: 11 + 13 = 24 17 + 19 = 36 23 + 29 =52 > 50 Hence 52 is answer
13	You are given an array of size N containing numbers from 1 to N. One number is **missing** and one number is **repeated exactly once**.

	Find the **repeating number** and the **missing number**. **Input**: N = 5 arr = [1, 2, 2, 4, 5] **Output**: Repeating = 2, Missing = 3
14	Array A is given in ascending order Array B is given in descending order Create a new array of n+m Containing all elements from both arrays in ascending order.